

7.7 Hyperbolic Functions

Calculus II

0.1 Hyperbolic Functions

0.1.1 Hyperbolic Functions

Hyperbolic Cosine

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad (1)$$

Hyperbolic Sine

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (2)$$

Hyperbolic Tangent

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3)$$

Hyperbolic Cotangent

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad (4)$$

Hyperbolic Secant

$$x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \quad (5)$$

Hyperbolic Cosecant

$$x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \quad (6)$$

0.1.2 Hyperbolic Identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \frac{1}{\cosh^2 x}$$

$$\coth^2 x - 1 = \frac{1}{\sinh^2 x}$$

$$\coth(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}$$

$$\cosh(-x) = \cosh x$$

$$\sinh(-x) = -\sinh x$$

$$\tanh(-x) = -\tanh x$$

$$\sinh 2x = 2 \sinh x \cosh x$$

$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$

0.1.3 Derivatives and Integral Formulas

$$1. \quad \frac{d}{dx}(\cosh x) = \sinh x \quad \Rightarrow \quad \int \sinh x \, dx = \cosh x + C$$

$$2. \quad \frac{d}{dx}(\sinh x) = \cosh x \quad \Rightarrow \quad \int \cosh x \, dx = \sinh x + C$$

$$3. \quad \frac{d}{dx}(\tanh x) = 1 - \tanh^2 x \quad \Rightarrow \quad \int \frac{1}{1 - \tanh^2 x} \, dx = \tanh x + C$$

$$4. \quad \frac{d}{dx}(\coth x) = -\frac{1}{\sinh^2 x} \quad \Rightarrow \quad \int \frac{1}{\sinh^2 x} \, dx = -\coth x + C$$

$$5. \quad \frac{d}{dx}(x \tanh x) = \tanh x + x \operatorname{sech}^2 x \quad \Rightarrow \quad \int x \operatorname{sech}^2 x \, dx = x \tanh x - \ln |\cosh x| + C$$

$$6. \quad \frac{d}{dx}(x \coth x) = \coth x - x \operatorname{csch}^2 x \quad \Rightarrow \quad \int x \operatorname{csch}^2 x \, dx = x \coth x - \ln |\sinh x| + C$$

0.1.4 Integrals of Hyperbolic Functions

1. $\int \tanh x \, dx = \ln \cosh x + C$
2. $\int \coth x \, dx = \ln |\sinh x| + C$
3. $\int x \, dx = \tan^{-1} |\sinh x| + C$
4. $\int x \, dx = \ln |\tanh(x/2)| + C$

0.1.5 Inverses of the Hyperbolic Functions Expressed as Logarithms

$$\begin{aligned} \cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) \quad (x \geq 1) & {}^{-1}x &= \cosh^{-1} \frac{1}{x} \quad (0 < x \leq 1) \\ \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) & {}^{-1}x &= \sinh^{-1} \frac{1}{x} \quad (x \neq 0) \\ \tanh^{-1} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad (|x| < 1) & \coth^{-1} x &= \tanh^{-1} \frac{1}{x} \quad (|x| > 1) \end{aligned}$$

0.1.6 Derivatives of the Inverse Hyperbolic Functions

$$\begin{aligned} \frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2 - 1}} \quad (x > 1) & \frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{x^2 + 1}} \\ \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1 - x^2} \quad (|x| < 1) & \frac{d}{dx}(\coth^{-1} x) &= \frac{1}{1 - x^2} \quad (|x| > 1) \\ \frac{d}{dx}({}^{-1}x) &= -\frac{1}{x\sqrt{1 - x^2}} \quad (0 < x < 1) & \frac{d}{dx}({}^{-1}x) &= -\frac{1}{|x|\sqrt{1 + x^2}} \quad (x \neq 0) \end{aligned}$$

0.1.7 Integral Formulas

1. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a} + C$, for $x > a$
2. $\int \frac{dx}{x^2 + a^2} = \sinh^{-1} \frac{x}{a} + C$, for all x
3. $\int \frac{dx}{a^2 - x^2} = \frac{1}{a} \tanh^{-1} \frac{x}{a} + C$, for $|x| < a = \frac{1}{a} \coth^{-1} \frac{x}{a} + C$, for $|x| > a$
4. $\int \frac{dx}{x\sqrt{a^2 - x^2}} = -\frac{1}{a} {}^{-1} \frac{x}{a} + C$, for $0 < x < a$
5. $\int \frac{dx}{x\sqrt{a^2 + x^2}} = -\frac{1}{a} {}^{-1} \frac{|x|}{a} + C$, for $x \neq 0$